# An autonomous career system for embodied creative research and development

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**Autonomous agents are increasingly able to plan, monitor, and execute tool-mediated workflows, but a career objective is normally treated as motivation rather than as a measurable control problem. We report an autonomous career system that converts a target role, Walt Disney Imagineering Research and Development mechanical Imagineering in Glendale, into a logged state machine with role-fit dimensions, guardrails, review cycles, continuous manuscript updates, and public artifacts. The review loop uses GPT-5.5 and a whole-public-AO-Labs source graph, including the CV, Sarrus, FluxCell, Relay, Relay Live, Progress, Curtis, Ocean, Talk, Yum, Lily, other public project surfaces, the WDI R&D profile, the active WDI listing, and Disney Research context. It can critique public sources autonomously but cannot send outreach, claim relationships, submit applications, or use private email/contact data without approval. At the current lock on 9 May 2026, the production system tracked six role-fit dimensions, four portfolio anchors, one daily cycle, fourteen AI reviews, zero outreach events, five public artifacts, and twenty-four journal entries. The live dashboard now exposes compact operating facts rather than explanatory coaching copy: paper state, WDI R&D state, mechanical signal, principal signal, system source count, profile freshness, approval boundary, outcome signals, and backend status. The live fit signal is 74 of 100 with a credible-but-needs-signal confidence label. The latest autonomous review scored the public profile 76 of 100 from 28 sources and identified the same central constraint: Sarrus makes the active Glendale mechanical-design rung credible, while principal-scope ownership and an active principal posting remain unverified. On 7 May 2026, the CV was revised to remove the redundant Sarrus subpage split and point the public record toward mechanical R&D, soft robotics, motion systems, and human-facing physical prototypes. A later same-day revision removed the named-role phrase from the CV so the document reads as a serious mechanical R&D record rather than a named-company pitch. The**

**same day, `https://progress.aolabs.io` was introduced as a shared AO Labs progress ledger that scans public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state into persistent longitudinal snapshots. On 9 May 2026, the profile page added an explicit freshness timestamp and widened the source graph so general AO Labs work is counted as profile context while role-fit credit remains bounded by direct relevance to mechanical R&D, physical interaction, prototypes, systems, and principal-scope ownership. The paper itself is part of the control loop. Any substantive change to a paper-backed AO Labs system must update the manuscript and PDF so the public artifact remains a cumulative record rather than a stale explainer. The contribution is therefore not a claim that the system has secured a job. It is a falsifiable protocol for making an otherwise ambiguous career-conversion process auditable, adaptive, continuously documented, and constrained by truthful public sources.**

# 1 Introduction

Tool-using language-model systems can now reason, call external services, write artifacts, and update deployed software (*1*–*4*). These capabilities create an opportunity to treat long-horizon personal objectives as monitored systems rather than as episodic intentions, in the older spirit of autonomic systems and closed-loop discovery workflows (*5*, *6*). A career target is especially difficult because the desired outcome depends on evidence quality, technical fit, social trust, timing, applications, and human judgment. Without a state model, repeated effort can feel intense while remaining scientifically uninterpretable.

This paper describes an early deployment of an autonomous career-signal system aimed at WDI Research and Development mechanical Imagineering roles. The system uses the active WDI Research and Development Imagineer - Mechanical Design Engineer role as the live rung and the Principal R&D Imagineer - Mechanical Engineer profile as the north-star target (*7*). The positioning line is deliberately stable: mechanical PhD, soft robotics, creative prototyping, and AI-assisted tools for physical interaction systems.

The design borrows from controlled revenue experimentation: define a measurable state, choose the current bottleneck, constrain allowed actions, record interventions, and update a paper from real source material rather than from aspiration. The system is not allowed to invent credentials, inflate relationships, send spam, or apply on behalf of the operator. It can monitor state, update dashboards, maintain a paper, identify the next constraint, run a source-bounded AI review, and create or improve public artifacts that are grounded in existing work. The paper is not a one-time publication draft. It is a cumulative methods record, and it must change whenever the system changes in a way that affects evidence, policy, interface, metrics, sources, guardrails, or interpretation.

# 2 System architecture

The deployment consists of a public dashboard, a Railway-hosted backend, a JSON runtime state file mounted on a persistent volume, and scheduled or operator-triggered automation. The backend exposes health, operations-check, journal, weekly-paper, event, daily-cycle, and AI-review endpoints. The dashboard is intentionally direct. Its current sections are: paper artifact, WDI R&D state, compact signal strip, outcome signals, and autonomous-loop status. The signal strip reports mechanical score, principal score, source count, profile freshness, and approval boundary without second-person coaching

**Table 1.** State variables at the current evidence lock. Values are reported from the live system after the profile-freshness and compact-readout update on 9 May 2026.

| Variable | Recorded value |
| --- | --- |
| Target company and location | Walt Disney Imagineering R&D, Glendale, California |
| Live rung | WDI Research and Development Imagineer - Mechanical Design Engineer |
| North star | Principal R&D Imagineer - Mechanical Engineer |
| Fit signal | 74 of 100 with credible-but-needs-signal confidence |
| Dominant bottleneck | Principal signal: visible ownership, technical direction, source depth, and approval-gated external validation |
| Counters | One daily cycle; fourteen AI reviews; zero outreach events; four portfolio anchors; five public artifacts; twenty-four journal entries |
| Active experiment | Autonomous career loop v0 |
| Reviewer source graph | Twenty-eight sources in the latest production run; the explicit source registry now includes Progress, Curtis, Relay Live, Yum, Lily, papers, dashboards, and discovered AO Labs links |
| Human approval boundary | Applications, sensitive outreach, referrals, claims involving real people, and any private contact/email data |
| Profile freshness | Public profile exposes the latest profile-state timestamp and source-count readout |
| Paper continuity rule | Any substantive change to a paper-backed system updates the manuscript and PDF in the same work cycle |

or explanatory filler.

The reviewer is a separate loop rather than a human-contact substitute. It builds a source bundle from the live state, the target listing, the WDI R&D profile, the paper PDF, the CV, Sarrus, FluxCell, Relay, Relay Live, Progress, Curtis, Ocean, Talk, Nerve, Duet, Violin, Yum, Lily, AO Labs home, Disney Research roster, and a Disney Research bipedal-character paper page. It also includes a compact identity profile summarizing the operator as a mechanical engineering PhD candidate building soft robotic materials, reconfigurable mechanisms, motion prototypes, human-facing physical systems, public project surfaces, research papers, and autonomous systems. The model returns source-bounded JSON: verdict, score, current constraint, signals, open limitations, candidate system work, and approval boundaries. If the model fails or times out, a deterministic fallback reviewer produces a conservative critique and records the fallback reason.

## 3 Decision policy

The policy scores six role-fit dimensions: mechanical depth, creative prototyping, human-facing motion, principal signal, Glendale application profile, and the autonomous system itself. Each dimension reports the tracked signal, basis, and source-backed limitation that would move the score. The score basis is intentionally calibrated: a base prior from the state file, a bounded event bonus, a bounded portfolio bonus, a daily-cycle bonus, optional review-loop credit, and a readiness ceiling that names the unresolved blocker. A score of 100 is reserved for no known blocker in that lane, not for accumulated activity. In the current lock, the weakest live signal is principal signal. The current internal work is to make visible ownership, source depth, and technical direction stronger without sending outreach before approval.

This policy is intentionally conservative. It can make public artifacts, update the dashboard, refresh the manuscript, run autonomous critique, and record state. It cannot manufacture the social trust that a principal-level role requires. If an action requires a real person's attention or reputation, the system must stop at preparation and request approval before sending. The review loop therefore becomes a repeatable calibration instrument, not a license to automate human outreach.

## 4 Results

The current deployment is functional and has moved beyond the initial dashboard-only state. The dashboard and backend are live, the paper endpoint can publish a snapshot, the site exposes a PDF manuscript, and the AI reviewer has produced live production critiques from the public source graph. The latest reviewer run used GPT-5.5, consumed 28 sources, returned a review score of 76 of 100, and identified a narrow dominant issue: the strongest public record supports the active Glendale WDI Research and Development Imagineer - Mechanical Design Engineer rung, while the Principal R&D Imagineer north-star remains unverified.

The largest artifact change in this cycle was the Sarrus technical record. The system published geometry, compliant-link and flexural-joint path, pressure-driven expansion, module assembly, build state, force/stiffness figures, and motion examples. It then added an HTML-visible measurement table with pressure range, expansion ratio, estimated leg angle, output force, local stiffness, hysteresis status, planar dome height, double-dome resolution, and prototype limitation. The WDI R&D profile now uses this table as the dominant technical path while preserving the current data boundary: raw digitized force, stiffness, and hysteresis curves with uncertainty remain unresolved.

After the latest interface revision, the live site uses a compact operating surface instead of explanatory tiles or second-person coaching. The state surface now reports compact facts: Mechanical 86/100 with geometry, actuation, assembly, measurement, and motion; Principal 56/100 with ownership, technical direction, collaborators, and validation; System with 28 sources; profile freshness; and Boundary with applications, referrals, direct outreach, and person-facing claims. The type scale was reduced because the dashboard is an operating surface, not a poster. The outcome-signal cards use the same standard: direct signal, current source state, and system-owned operation. This is a functional requirement, not a tone preference.

The system-level fit signal is now calibrated to 74 of 100 after removing the earlier saturation behavior that let logged work push several lanes to 100. This should be interpreted as an internal readiness signal, not as an external hiring probability. The review result is deliberately separate: it scores the current public profile, not the probability of being hired. It confirms that the overall direction is credible for WDI R&D mechanical design while rejecting the idea that the current public profile is already principal-level. The profile now includes both a principal-scope boundary and an above-fold role-verification banner: the active target is the Glendale mechanical-design rung; the principal title remains a north-star profile until an active principal posting and principal-scope leadership work are independently verified.

The CV was revised in the same paper cycle because it is part of the reviewer source graph and the public application record. The selected-public-work section now treats Sarrus as one coherent project rather than splitting it into a separate subpage link. The top line and profile now point toward mechanical R&D, soft robotics, physical interaction, motion systems, and human-facing physical prototypes without naming the target directly. The Sarrus research and publication entries now use Sarrus-linkage soft-robotic cells and planar/cylindrical/locomotion assemblies instead of a generic topology frame. This keeps the CV aligned with the WDI R&D objective without reading like a named-company pitch or overstating principal-scope evidence.

The progress ledger adds a cross-application memory layer rather than replacing the Imagineer dashboard. Its first deployed version scans eighteen public sources, including AO Labs home, the CV PDF, Sarrus, FluxCell, the Imagineer dashboard, the Imagineer API state, the Imagineer paper PDF, Curtis home, Curtis media state, Relay home, Relay backend health, Relay intent state, Talk, Ocean, Nerve, Duet, Lily, and Yum. The initial production snapshot reported eighteen healthy sources, one persistent snapshot on the Railway volume, an Imagineer fit score of 74, and 207 Curtis practice-media

**Table 2.** Latest autonomous review result. The reviewer is a GPT-5.5 source-bounded critique over the public AO Labs graph, not a human recommendation.

| **Field** | **Recorded value** |
|---|---|
| Model | GPT-5.5 via the Responses API |
| Scope | Whole public AO Labs graph plus WDI role and Disney Research context |
| Source count | 28 sources in the latest production run |
| Review score | 76 of 100 |
| Verdict | Credible for the active WDI Research and Development Imagineer - Mechanical Design Engineer target; not yet calibrated for a Principal R&D Imagineer mechanical role |
| Top issue | Principal-scope ownership is not visible enough: technical direction, led design reviews, mentorship, budget ownership, schedule ownership, shipped systems, or integrated subsystem ownership remain unverified |
| Current system work | Direct state surface, Sarrus-first profile, source-backed role boundary, continuous paper update |

inventory items. This creates the storage surface needed for long-term trend comparisons across papers, practice, portfolio state, public pages, and outcome systems.

The profile page now exposes its own update timestamp. This matters because the profile is not only a Disney-specific pitch; it is a compressed read of the public AO Labs record. New systems such as Curtis, unrelated project surfaces such as Yum, updates to Sarrus, paper endpoints, and progress-ledger snapshots are counted as context about the operator's range, persistence, taste, and systems-building pattern. They do not automatically increase role-fit scores. The scoring boundary remains explicit: only source material that strengthens mechanical R&D, physical interaction, prototype execution, systems ownership, or principal-scope responsibility should move the Imagineer readiness state.

## 5 Limitations and guardrails

The system cannot guarantee an offer. A job decision depends on timing, hiring needs, competition, interview performance, portfolio interpretation, and institutional context. The useful claim is narrower: the system can make progress legible, choose the next bottleneck, prevent fake progress, and turn repeated effort into logged artifacts. It should be evaluated by public artifact velocity, critique quality, application readiness, interviews, and eventual conversion.

The guardrails are part of the result. The system forbids fabricated credentials, fake outreach, invented

**Table 3.** Role-fit dimensions after the current autonomous review cycle. Scores are internal operating metrics, not external hiring probabilities; the live dashboard exposes the tracked signal, basis, and unresolved signal for each row.

| Dimension | Score | Current interpretation |
|---|---|---|
| Mechanical depth | 86 | Strong but not complete; Sarrus has a public technical record and measurement table, with raw digitized curves, uncertainty, and test conditions still pending. |
| Creative prototyping | 82 | Strongest signal; should remain attached to concrete physical demonstrations. |
| Human-facing motion | 79 | Credible signal, but still needs a concise guest-facing motion sequence. |
| Principal signal | 56 | Current bottleneck; visible ownership, technical direction, collaborators, and external validation remain thin. |
| Glendale profile | 72 | Credible for the active rung; CV framing is now aligned, while demo reel and role-specific narrative remain open. |
| Autonomous system | 72 | Backend, dashboard, durable state, reviewer, journal, and paper loop are live, but scheduled runs, evaluation history, and outcome tracking are early. |

Disney relationships, automated applications, and claims that are not supported by public or logged sources. Private email/contact data is excluded from the reviewer unless explicitly approved. These constraints reduce short-term aggressiveness, but they preserve interpretability. If the system ever reports strong progress, that progress should be traceable to artifacts, logs, AI reviews, human approvals, applications, or outcomes.

## 6 Availability

The live dashboard is available at `https://imagineer.aolabs.io/`, with a fallback at `https://aolabs.io/imagineer/`. The live backend exposes `https://imagineer.aolabs.io/api/imagineer/ops-check`, `https://imagineer.aolabs.io/api/imagineer/weekly-paper`, and `https://imagineer.aolabs.io/api/imagineer/ai-review`. This manuscript is a public artifact and must be refreshed from logged sources in the same work cycle as substantive changes to the dashboard, backend, reviewer policy, source graph, public profile, Sarrus/portfolio evidence, metrics, or guardrails.

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